

MH2100 Calculus III

Midterm II (Solutions)

AY 2022/2023, Semester 1

Quantitative Research Society @NTU

October 2022

Overview

- **Historical paper:** Midterm II, October 2022; **90 minutes** were allowed.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted.
- **Original format:** three questions worth 50 marks in total. ONE two-sided A4 piece of paper with handwritten notes was permitted.
- **2026 scope note:** the current midterm scope ends at double integrals, including polar coordinates. Cylindrical and spherical coordinates, general Jacobian transformations, and line integrals are beyond that scope; these historical questions are retained and labelled accordingly.
- **Source:** MH2100 Midterm II, AY2022/2023 Semester 1. The instructions and questions below reproduce the official paper; solutions and marking schemes are unofficial QRS@NTU material.

Instructions to candidates (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates should write their answers in the separate answer book provided.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(16 Marks)****QUESTION 1.**

- (i) [5 marks] Find the *maximum* and *minimum* values of xy satisfying the condition that $3x^2 + 2y^2 = 36$. Also give the points (x, y) where the *maximum* and *minimum* values are achieved.
- (ii) [3 marks] Evaluate the iterated integral.

$$\int_{-1}^1 \int_0^1 2ye^x \, dx \, dy.$$

- (iii) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_0^{\sqrt{\pi/2}} \int_y^{\sqrt{\pi/2}} 2 \cos(x^2) \, dx \, dy.$$

- (iv) [4 marks] Find the area of the region D (in the xy -plane) bounded by the cardioid expressed in polar coordinates as

$$r = 1 - \sin \theta; \quad \text{and} \quad 0 \leq \theta \leq 2\pi.$$

Hint: the formula $\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$ may be useful.

Solution**Method 1: Direct computation**

- (i) $a = x^2$, $b = y^2 = \frac{36-3a}{2}$, $0 \leq a \leq 12$:

$$(xy)^2 = ab = 54 - \frac{3}{2}(a-6)^2 \leq 54 \implies |xy| \leq 3\sqrt{6} \quad (\text{equality iff } a = 6, b = 9)$$

$$\boxed{\max xy = 3\sqrt{6} \text{ at } (\sqrt{6}, 3), (-\sqrt{6}, -3),}$$

$$\boxed{\min xy = -3\sqrt{6} \text{ at } (\sqrt{6}, -3), (-\sqrt{6}, 3).}$$

- (ii)

$$\begin{aligned} \int_{-1}^1 \int_0^1 2ye^x \, dx \, dy &= (e-1) \int_{-1}^1 2y \, dy && (\text{inner integral}) \\ &= (e-1) [y^2]_{-1}^1 = \boxed{0} \end{aligned}$$

- (iii) $D : 0 \leq y \leq x \leq \sqrt{\pi/2}$; reverse the order (Fubini):

$$\begin{aligned} \int_0^{\sqrt{\pi/2}} \int_y^{\sqrt{\pi/2}} 2 \cos(x^2) \, dx \, dy &= \int_0^{\sqrt{\pi/2}} \int_0^x 2 \cos(x^2) \, dy \, dx \\ &= \int_0^{\sqrt{\pi/2}} 2x \cos(x^2) \, dx \\ &= [\sin(x^2)]_0^{\sqrt{\pi/2}} = \boxed{1} && (u = x^2) \end{aligned}$$

(iv)

$$A = \frac{1}{2} \int_0^{2\pi} r^2 d\theta = \frac{1}{2} \int_0^{2\pi} (1 - 2\sin\theta + \sin^2\theta) d\theta \quad (r = 1 - \sin\theta)$$

$A = \frac{1}{2}(2\pi + \pi) = \frac{3\pi}{2}.$

$$(\int \sin\theta = 0, \int \sin^2\theta = \pi)$$

Method 2: AM–GM

Part (i):

$$36 = 3x^2 + 2y^2 \geq 2\sqrt{6}|xy| \quad (\text{AM–GM})$$

$$\implies |xy| \leq 3\sqrt{6}, \text{ equality iff } 3x^2 = 2y^2 = 18 \iff x^2 = 6, y^2 = 9$$

$$\therefore \max xy = 3\sqrt{6} \text{ at } (\sqrt{6}, 3), (-\sqrt{6}, -3)$$

$$\min xy = -3\sqrt{6} \text{ at } (\sqrt{6}, -3), (-\sqrt{6}, 3)$$

Marking scheme (16 marks)(i) **(5 marks)**

- 2 marks: reduces the constraint to a correct one-variable expression for $(xy)^2$.
- 1 mark: maximises that expression at $x^2 = 6, y^2 = 9$.
- 2 marks: gives both extrema and all four attaining points.

(ii) **(3 marks)**

- 1 mark: evaluates the inner integral correctly.
- 1 mark: recognises the odd outer factor or completes the outer integral.
- 1 mark: obtains 0.

(iii) **(4 marks)**

- 2 marks: reverses the order with correct bounds.
- 1 mark: reduces to $\int 2x \cos(x^2) dx$.
- 1 mark: obtains 1.

(iv) **(4 marks)**

- 1 mark: states the polar area formula.
- 2 marks: expands and integrates the square correctly.
- 1 mark: obtains $3\pi/2$.

Question 2**(17 Marks)***(beyond 2026 midterm scope)***QUESTION 2.**

- (i)
- (beyond 2026 midterm scope)*
- [4 marks] Use cylindrical coordinates to evaluate

$$\iiint_E \sqrt{x^2 + y^2} \, dV,$$

where E is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 1$ and $z = 2$.

- (ii)
- (beyond 2026 midterm scope)*
- Consider the transformation
- $u = x + 4y$
- and
- $v = y - x$
- .

(a) [3 marks] Calculate the Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$.

- (b) [5 marks] Use the transformation to evaluate the iterated integral

$$\int_0^1 \int_y^{5-4y} (x + 4y)e^{y-x} \, dx \, dy.$$

Hint: you can use $\int x e^{ax} \, dx = \frac{e^{ax}(ax - 1)}{a^2} + C$.

- (iii)
- (beyond 2026 midterm scope)*
- [5 marks] Use spherical coordinates to find the volume of the region inside the sphere
- $x^2 + y^2 + z^2 = 4$
- and outside the cone
- $z^2 = x^2 + y^2$
- . That is, the volume of the region
- E
- given by

$$E = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 4 \text{ and } z^2 \leq x^2 + y^2\}.$$

Solution**Method 1: Direct computation**

- (i)
- $E : 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi, 1 \leq z \leq 2; \sqrt{x^2 + y^2} = r, \, dV = r \, dz \, dr \, d\theta:$

$$\iiint_E \sqrt{x^2 + y^2} \, dV = \int_0^{2\pi} \int_0^1 \int_1^2 r^2 \, dz \, dr \, d\theta = 2\pi \left[\frac{r^3}{3} \right]_0^1 = \boxed{2\pi/3}$$

(ii) (a) $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 4 \\ -1 & 1 \end{vmatrix} = \boxed{5}$

(b)

$$x = \frac{u - 4v}{5}, \quad y = \frac{u + v}{5}, \quad \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{5}$$

$$x = y, \, x + 4y = 5, \, y = 0 \mapsto v = 0, \, u = 5, \, v = -u$$

$$(0, 0), (5, 0), (1, 1) \mapsto (0, 0), (5, -5), (5, 0) \implies 0 \leq u \leq 5, \, -u \leq v \leq 0$$

$$\int_0^1 \int_y^{5-4y} (x + 4y)e^{y-x} \, dx \, dy = \frac{1}{5} \int_0^5 \int_{-u}^0 u e^v \, dv \, du$$

$$\begin{aligned}
&= \frac{1}{5} \int_0^5 u(1 - e^{-u}) \, du \\
&= \frac{1}{5} \left(\frac{25}{2} - 1 + 6e^{-5} \right) \quad \left(\int_0^5 u e^{-u} \, du = 1 - 6e^{-5} \right) \\
&= \boxed{\frac{23}{10} + \frac{6}{5e^5}}.
\end{aligned}$$

(iii) ϕ from the $+z$ -axis; $E : 0 \leq \rho \leq 2, \pi/4 \leq \phi \leq 3\pi/4, 0 \leq \theta \leq 2\pi$:

$$\begin{aligned}
V &= \int_0^{2\pi} \int_{\pi/4}^{3\pi/4} \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2\pi \cdot \frac{8}{3} \cdot [-\cos \phi]_{\pi/4}^{3\pi/4} \\
&= \boxed{V = \frac{16\sqrt{2}\pi}{3}}.
\end{aligned}$$

Marking scheme (17 marks)

- (i) **(4 marks)** 1 mark: correct cylindrical bounds; 1 mark: correct integrand and Jacobian; 1 mark: correct integration; 1 mark: obtains $2\pi/3$.
- (ii) **(8 marks)**
- (a) **(3 marks)** 2 marks: forms the determinant; 1 mark: obtains 5.
- (b) **(5 marks)** 1 mark: inverts the transformation and finds $|\partial(x, y)/\partial(u, v)|$; 2 marks: maps the region correctly; 1 mark: transforms the integral; 1 mark: obtains $23/10 + 6/(5e^5)$.
- (iii) **(5 marks)** 2 marks: correct spherical bounds; 2 marks: correct volume integral; 1 mark: obtains $16\sqrt{2}\pi/3$.

Question 3**(17 Marks)***(beyond 2026 midterm scope)***QUESTION 3.**

- (i) *(beyond 2026 midterm scope)* [4 marks] Let C be the curve parametrised by $x = 3t$ and $y = t^3$ where $0 \leq t \leq 1$. Evaluate $\int_C (x^3 + y) \, ds$.
- (ii) *(beyond 2026 midterm scope)* [4 marks] Find the work done by the vector field $\mathbf{F} = \langle x^3 - y^3, xy^2 \rangle$ along the curve parametrised by $x = t^2$ and $y = t^3$ from $t = -1$ to $t = 0$.
- (iii) *(beyond 2026 midterm scope)* [4 marks] Determine if the vector field $\mathbf{F} = \langle 3x^2y^2 + e^x, 2x^3y + \sin(y) \rangle$ is conservative. If so, find a potential function f for $\mathbf{F}$.
- (iv) *(beyond 2026 midterm scope)* [5 marks] Let C be part of the curve (in the xy -plane) described by polar coordinates as $r = \sin(\theta)$ where $0 \leq \theta \leq \pi$. Find the length of C .

Solution**Method 1: Direct computation**

- (i) $x' = 3$, $y' = 3t^2$, so $ds = 3\sqrt{1+t^4} \, dt$; $x^3 + y = 27t^3 + t^3 = 28t^3$:

$$\begin{aligned} \int_C (x^3 + y) \, ds &= 84 \int_0^1 t^3 \sqrt{1+t^4} \, dt = 21 \int_1^2 u^{1/2} \, du \quad (u = 1+t^4) \\ &= \boxed{14(2\sqrt{2} - 1)} \end{aligned}$$

- (ii) $\mathbf{r}' = \langle 2t, 3t^2 \rangle$, $\mathbf{F}(\mathbf{r}(t)) = \langle t^6 - t^9, t^8 \rangle$:

$$\int_{-1}^0 \mathbf{F} \cdot \mathbf{r}' \, dt = \int_{-1}^0 (2t^7 + t^{10}) \, dt = \left[\frac{t^8}{4} + \frac{t^{11}}{11} \right]_{-1}^0 = \boxed{-7/44}$$

- (iii) $\mathbf{F} = \langle P, Q \rangle$, $P_y = 6x^2y = Q_x$ on $\mathbb{R}^2 \implies \mathbf{F}$ conservative.

$$\begin{aligned} f &= \int P \, dx = x^3y^2 + e^x + h(y) \\ f_y &= 2x^3y + h'(y) = Q \implies h' = \sin y \implies h = -\cos y \\ &\boxed{f(x, y) = x^3y^2 + e^x - \cos y.} \end{aligned}$$

- (iv) $r = \sin \theta$, $r' = \cos \theta$:

$$L = \int_0^\pi \sqrt{r^2 + r'^2} \, d\theta = \int_0^\pi \sqrt{\sin^2 \theta + \cos^2 \theta} \, d\theta = \boxed{L = \pi}$$

Method 2: circle circumference

Part (iv):

$$r = \sin \theta \iff x^2 + (y - \tfrac{1}{2})^2 = \tfrac{1}{4}, \text{ traced once on } 0 \leq \theta \leq \pi \implies L = 2\pi \cdot \tfrac{1}{2} = \pi$$

Marking scheme (17 marks)

- (i) **(4 marks)** 1 mark: obtains ds ; 1 mark: substitutes the integrand; 1 mark: uses a valid substitution; 1 mark: obtains $14(2\sqrt{2} - 1)$.
- (ii) **(4 marks)** 1 mark: finds $\mathbf{r}'$; 2 marks: forms the work integrand; 1 mark: obtains $-7/44$.
- (iii) **(4 marks)** 1 mark: checks $P_y = Q_x$; 2 marks: constructs a potential; 1 mark: gives a valid final potential.
- (iv) **(5 marks)** 2 marks: states the polar arc-length formula; 2 marks: simplifies its integrand; 1 mark: obtains π .